

STA 1023/MAT 1063

CHAPTER 3

PROBABILITY DISTRIBUTIONS

3.1 Introduction

Definition 3.1 – Random Variable

If S is a sample space with a probability measure and X is a real valued function over the elements of S , then X is called a random variable.

Random Variable (R.V.) = X

The values that can be assigned to $X = x$

Example 3.1: Rolling a balance six-sided dice

Solution:

Example 3.2:

In a sample of 100 students, 25 students had no siblings, 35 students had one sibling, 20 students had two siblings, 12 students had three siblings, and the rest had four siblings. List the elements in the sample space and corresponding probabilities if the random variable X is the number of siblings.

3.2 Discrete Random Variables

Variables defined over discrete sample space are known as discrete random variables

Definition 3.2 – Probability Mass Function

If X is a discrete random variable, the function given by $P(x) = P(X = x)$ for each x within the range of X is called the probability mass function

(pmf).

Theorem 3.1

A function can serve as the probability distribution of discrete random variable X if and only if its value $P(x)$ satisfy the conditions,

- $P(x) \geq 0$ for each value within its domain
- $\sum_x P(x) = 1$

Example 3.2: revisited

- Graphical representation

Definition 3.3 – Cumulative Density Function; CDF

If X is a discrete random variable, the function given by,

$$F(x) = P(X \leq x) = \sum_{t \leq x} P(t) \text{ for } -\infty < x < \infty$$

is called Cumulative Density of X .

Example 3.2: revisited – Find the CDF of X

Example 3.3

Check whether the function given by $p(x) = \frac{x+2}{25}$ for $x = 1, 2, 3, 4, 5$ serve as a probability mass function of a discrete random variable and find the probabilities for the followings

a) $2 \leq P(x) \leq 4$

b) $P(x) < 2$

c) $P(x) \geq 2$

$$p(x) = \frac{x+2}{25} \text{ for } x = 1, 2, 3, 4, 5$$

$$\text{a) } 2 \leq P(x) \leq 4$$

$$\text{b) } P(x) < 2$$

$$\text{c) } P(x) \geq 2$$

CDF of X

Theorem 3.2

The values $F(x)$ of the distribution function of a discrete random variable X satisfy the conditions:

i) $F(-\infty) = 0$ and $F(+\infty) = 1$

ii) If $a < b$, then $F(a) \leq F(b)$ for any real numbers a and b .

Theorem 3.3

If the range of a random variable X consists of values $x_1 < x_2 < \dots < x_{n_1}$. Then,

$$P(x_1) = F(x_1) \text{ and}$$

$$P(x_i) = F(x_i) - F(x_{i-1}) \text{ for } i = 2, 3, \dots, n$$

Example 3.4

Suppose the cumulative density function (CDF) of a random variable is as follows.

$$F(x) = \begin{cases} 0 & ; -\infty < x < 0 \\ 1/16 & ; 0 \leq x < 1 \\ 5/16 & ; 1 \leq x < 2 \\ 11/16 & ; 2 \leq x < 3 \\ 15/16 & ; 3 \leq x < 4 \\ 1 & ; x \geq 4 \end{cases}$$

a) Find the pmf

b) $P(X = 2)$

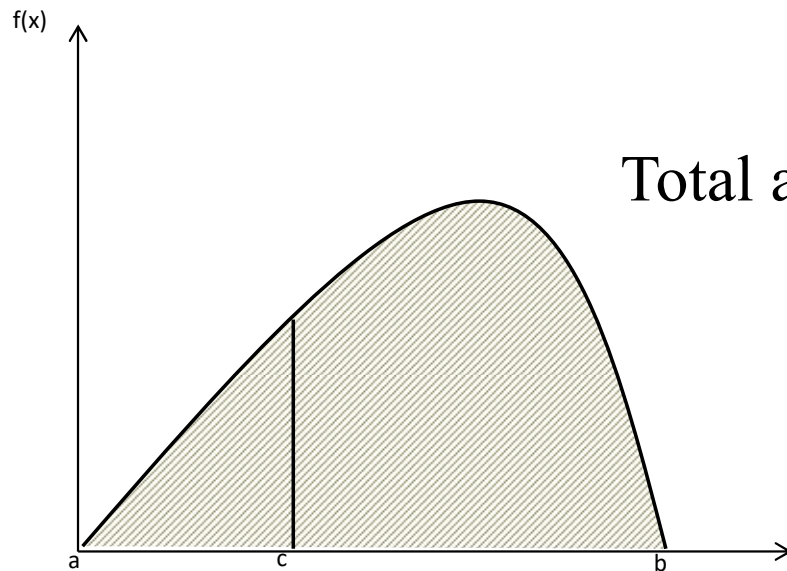
c) $P(X \geq 3)$

d) $P(2 \leq X \leq 4)$

3.3 Continuous Random Variables

Continuous random variables can have all the values within the given interval.

- Example: Temperature in Celsius – $16.5^{\circ}\text{C} \leq x \leq 32.5^{\circ}\text{C}$



Total area under the curve = Total probability = 1

$$\Rightarrow \int_a^b f(x) dx = 1$$

Definition 3.4 – The Probability Density Function; PDF

A function with values $f(x)$, defined over the set of all real numbers is called a pdf of the continuous random variable X if and only if,

$$P(a \leq x \leq b) = \int_a^b f(x) d(x) ;$$

for any real constant a and b , $a \leq b$

For continuous random variables, point probability cannot be found.

$$\int_c^c f(x) d(x) = 0$$

Theorem 3.4

If x is a continuous random variable and a and b are real constants with $a \leq b$,

$$P(a \leq x \leq b) = P(a < x \leq b) = P(a \leq x < b) = P(a < x < b)$$

Theorem 3.5

A function can serve as a probability density of a continuous random variable if its values satisfy the conditions,

1. $f(x) \geq 0$ for $-\infty < x < \infty$
2. $\int_{-\infty}^{+\infty} f(x) dx = 1 \quad \forall x \in R$

Example 3.5

Find the values of K so that $f(x)$ is a valid pdf of continuous random variable X where,

$$f(x) = \begin{cases} Kx^2 & ; 0 \leq x \leq 2 \\ 0 & \text{o/w} \end{cases}$$

Find

a) $P(0 \leq x \leq 1)$

b) $P(1 \leq x \leq 2)$

$$f(x) = \begin{cases} Kx^2 & ; 0 \leq x \leq 2 \\ 0 & o/w \end{cases}$$

Example 3.6

Find the value of K for a valid pdf

$$f(x) = \begin{cases} Ke^{-3x} & ; x > 0 \\ 0 & ; 0/w \end{cases}$$

Then find

a) $P(0.5 \leq x \leq 1)$ b) $P(1 \leq x \leq 5)$

